Abstract

Poisson regression is widely used for count outcomes, but covariates are frequently observed with classical measurement error, and the resulting naive estimates are attenuated with invalid confidence intervals. We develop Adaptive SIMEX (A-SIMEX), which automates the two ad-hoc choices of SIMEX—the extrapolation region and the polynomial degree—by cross-validation with a one-standard-error parsimony rule, adds a ridge-corrected second extrapolation stage, and is accompanied by complete asymptotic theory: consistency under a growing-degree regime that we make explicit; asymptotic normality with an exact decomposition of the asymptotic variance into data and simulation components (the latter vanishing at rate B^{-1} in the number of simulation replicates); and efficiency results relative to the oracle estimator, including an information-monotonicity lemma showing that no estimator based on noisy covariates can improve on the oracle to first order, while the second-order efficiency comparison is signed neither way. The theory corrects several claims of earlier SIMEX treatments and resolves the tension between consistency (which requires the polynomial degree to grow like $\log n$) and inference through an explicit window condition on the degree growth rate. In simulation studies with 1,000 Monte Carlo repetitions per design cell—including misspecified (Laplace) error, negative-binomial outcomes, error variance estimated from replicated measures, and corrected-score and regression-calibration competitors—A-SIMEX removes most of the naive attenuation with valid inference when the error is moderate, and we identify an important limitation of automatic tuning: held-out prediction on the observed scale systematically under-extrapolates when the error is large, because its population minimizer is the naive pseudo-parameter. We characterize the resulting bias–variance trade-off, quantify the efficiency premium of extrapolation against the oracle, and give practical remedies. The methods are implemented in an open-source R package with an analytic sandwich variance estimator that we validate against the parametric bootstrap, and are applied to NHANES 2017–2018, where the measurement error is directly measurable: self-reported weight in a model for the count of prescription medications. All results are reproducible from a documented archive with fixed seeds.

0.1 This Paper's Contributions

We address these gaps by developing Adaptive SIMEX (A-SIMEX) and by subjecting it to a level of scrutiny that we argue should be standard for bias-correction methods. The contributions are:

1. **Automated hyperparameter selection, with its limits quantified.** Extrapolation region and polynomial degree are selected by k-fold cross-validation of the held-out Poisson deviance, with a one-standard-error parsimony rule. This removes subjectivity and is fully reproducible. Our simulations, however, reveal a structural limitation that we do not paper over: the population minimizer of the observed-scale CV criterion is the naive pseudo-parameter $\beta(1)$, not β_0, so CV systematically under-extrapolates when the measurement error is large (Section **??**). Automatic tuning is a solution for small-to-moderate error and a documented failure mode for large error, for which we propose deconvolution-aware remedies.

2. **A bias-corrected two-stage extrapolation and an exact variance estimator.** The primary polynomial fit is stabilized by a ridge second stage, and inference uses an analytic sandwich that decomposes the asymptotic variance exactly into a *data* component and a *simulation* component of order B^{-1}; this decomposition corrects the additive "parameter-plus-extrapolation" split that appears in earlier SIMEX treatments and in a draft of this paper, and it extends robustly to overdispersed outcomes and to estimated error variance (Sections **??** and **??**).

3. **Complete and corrected asymptotic theory.** We prove consistency and asymptotic normality of the A-SIMEX estimator, a growing-degree central limit theorem that reconciles the

$\log n$ degree growth needed for consistency with centered inference through an explicit window condition on the analyticity margin, and oracle-efficiency results comprising an information-monotonicity lemma (no estimator based on noisy covariates beats the oracle to first order) and an explicit second-order expansion whose sign is design-dependent. Along the way we correct three claims that circulate informally for SIMEX: the exactness of the extrapolation target, the form of the variance decomposition, and the sign of the second-order efficiency loss.

4. **Evidence at replication standard.** All simulations (1,000 Monte Carlo repetitions per cell) and the application are generated by a single seed-fixed pipeline; the paper reports what the pipeline produces, including results that are unflattering to the method.

0.2 Roadmap

Section **??** reviews the model and standard SIMEX. Section **??** presents A-SIMEX: cross-validated selection of $(\lambda_{\max}, \text{degree})$ (Section 3.1), the ridge second stage (Section 3.2), and the analytic sandwich variance estimator (Section 3.3). Section **??** states and proves the main results: the population SIMEX curve and its analyticity (Lemmas on the pseudo-true parameter), the consistency theorem, the asymptotic normality theorem with the exact data/simulation variance decomposition, the growing-degree central limit theorem, and the oracle-efficiency theorem with the information-monotonicity lemma. Section **??** reports the simulation studies (1,000 repetitions per cell): the low-dimensional strong-signal scenario with n- and σ_ϵ^2-grids, misspecified Laplace error, computation times up to $p = 50$, negative-binomial outcomes, and estimated error variance from replicated measures. Section **??** applies the methods to NHANES 2017–2018, where self-reported weight is the error-prone covariate and the error variance is measurable within the same survey, with a sensitivity analysis over σ_ϵ^2. Section **??** describes the R package. Section 1 discusses limitations and extensions. Complete proofs are collected in a companion appendix (reproduced in the replication archive as `asimex_proofs.pdf`), which supersedes the proof sketches of the earlier draft.

0.3 Variance Estimation and Confidence Intervals

Let $c = (c_0, \ldots, c_{K-1})$ denote the combined extrapolation weights (primary polynomial plus ridge second stage), so that the A-SIMEX estimator is exactly the linear functional $\hat\beta_{\text{A-SIMEX}} = \sum_j c_j \bar\beta(\lambda_j)$ with $\sum_j c_j = 1$, and let $\boldsymbol{H}_j = \mathbb{E}\big[e^{\beta(\lambda_j)^\top W(\lambda_j)} W(\lambda_j) W(\lambda_j)^\top\big]$ be the per-observation Hessian of the population SIMEX objective at grid point λ_j. Under the conditions of the asymptotic normality theorem,

$$\operatorname{Var}\big(\hat\beta_{\text{A-SIMEX}}\big) \;\approx\; \frac{1}{n}\boldsymbol{C}\big(\boldsymbol{V}^{\text{data}} + \tfrac{1}{B}\boldsymbol{V}^{\text{sim}}\big)\boldsymbol{C}^\top, \qquad \boldsymbol{C} = \big[c_0\boldsymbol{H}_0^{-1}, \ldots, c_{K-1}\boldsymbol{H}_{K-1}^{-1}\big], \tag{1}$$

where $\boldsymbol{V}^{\text{data}}$ is the covariance across observations of the stacked conditional influence functions $\mathbb{E}[\boldsymbol{m}_{ib}(\lambda_j) \mid Y_i, \boldsymbol{W}_i]$ and $\boldsymbol{V}^{\text{sim}}$ their conditional covariance given the data. Both matrices admit closed-form expressions under the Gaussian error model (Gaussian tilt identities) and are estimated from the same replicates that produce the SIMEX curve, at negligible additional cost; a pairs-bootstrap option is provided for small samples. Two features distinguish (1) from the earlier draft's decomposition $\text{Var}_{\text{param}} + \text{Var}_{\text{extrap}} + \text{Var}_{\text{sim}}$ with $\text{Var}_{\text{param}} = \boldsymbol{I}_W(\boldsymbol{\beta}_0)^{-1}$: (i) the naive Fisher information $\boldsymbol{I}_W(\boldsymbol{\beta}_0)^{-1}$ does *not* enter additively — the estimator is a fixed weighted average of the grid-point pseudo-estimators, and its variance is the propagated influence covariance (1), which reduces to the naive variance only in the limit of perfectly correlated grid-point influence; and (ii) the simulation noise enters at exactly $O(B^{-1})$, so $B = 100$ makes it negligible relative to the data component — confirmed empirically in Section **??**, where the analytic sandwich recovers 66–91% of the parametric-bootstrap variance (first-order under-statement, improving with n). Confidence intervals are $\hat{\beta}_{\text{A-SIMEX},j} \pm z_{1-\alpha/2}\,\widehat{\text{se}}_j$ with $\widehat{\text{se}}$ from (1); under overdispersion the same sandwich is valid with robust scores, and Section **??** verifies this under negative-binomial outcomes.

0.4 Computational Complexity

Three implementation ideas reduce the cost far below naive estimates. First, all B SIMEX replicates at a grid point are fitted by one *batched* IRLS sweep: the score and Hessian accumulations are tensor contractions over a (B, n, p) array, so the B fits share the linear-algebra machinery. Second, the CV procedure exploits the *prefix property* of the SIMEX grid: one union-grid curve per fold (with noise shared across λ within a replicate, per the manuscript's Algorithm 1) serves all candidate $\lambda_{\max}$, reducing the number of fitted pseudo-samples per fold from $|\mathcal{K}_\lambda| \cdot |\Lambda|\, B_{\text{cv}}$ to $|\Lambda|\, B_{\text{cv}}$. Third, fits along the curve are warm started from the previous grid point. Measured single-core times for the full A-SIMEX fit (CV plus final curve plus sandwich) are 0.10, 0.28, 0.95, and 3.3 seconds at $(n{=}200, p) \in \{5, 15, 30, 50\}$ with $B = 100$, $B_{\text{cv}} = 25$; standard SIMEX is about twice as fast, and the corrected-score estimator is one to two orders of magnitude cheaper but numerically fragile (Section **??**). The asymptotic complexity per replication remains $O(B\,|\Lambda|\,np^2)$ for the final curve plus $O(5\,|\Lambda|\,B_{\text{cv}}\,np^2)$ for the CV, but the constants are small enough that parallelization over Monte Carlo repetitions, CV folds, or grid points—all embarrassingly parallel, and all supported by the package through the `future`/`furrr` backends—is a matter of convenience rather than necessity on modern hardware.

1 Discussion

1.1 Summary of Contributions

This paper makes SIMEX for Poisson regression automatic, gives it complete and corrected asymptotic theory, and reports its finite-sample behavior at replication standard. Three findings that we believe are of independent interest emerged from holding the method to this standard: (i) held-out prediction on the observed scale cannot select the extrapolation that bias correction requires, because its population minimizer is the naive pseudo-parameter; (ii) the variance of the extrapolated functional is governed by the norm of the extrapolation weights ($\sum_j |c_j| \approx 7.8$ for the standard quadratic extrapolant), which quantifies a substantial and previously unappreciated efficiency premium; and (iii) the analytic data-plus-simulation sandwich makes inference nearly free, and its 66–91% recovery of the bootstrap variance delineates what part of the finite-sample noncoverage is variance miscalibration and what part is residual extrapolation bias.

1.2 Comparison with Alternatives

Versus standard SIMEX. A-SIMEX removes the ad-hoc choices and adds the variance machinery; in the simulations the two track each other closely, with A-SIMEX (1-SE rule) marginally better at small error and marginally worse at large error, where the parsimony rule reinforces the CV's under-extrapolation. The honest summary is that the tuning layer is not where the accuracy is won or lost; the extrapolation bias is.

Versus the corrected score (Nakamura, 1990). The corrected score is consistent for the Gaussian-error pseudo-score without requiring a model for $\Pr(X \mid W)$ — it requires only the error distribution, as SIMEX does — and is dramatically cheaper (< 0.05 s at $p = 50$). Its weaknesses, quantified in Section **??**, are numerical: Newton divergence in about 1% of fits under misspecification, RMSE exploding with p when many coefficients are weakly identified (0.13 at $p = 5$ to 2.72 at $p = 50$), and instability at small n with large error. SIMEX-type methods are more stable in exactly these regimes, at a computational premium. The two are complements: the corrected score when p is large and the error small, SIMEX when stability matters.

Versus deconvolution and functional methods. These remain computationally intensive and ill-posed; A-SIMEX scales to $p = 50$ in seconds. A fair comparison at the level of Delaigle and Meister (2008) would be valuable but is beyond our scope.

1.3 Limitations and Caveats

1. **Error variance known or estimated.** The theory conditions on $\boldsymbol{\Sigma}_\epsilon$. The replicate-based estimator performs indistinguishably in Section **??**; the estimated-variance asymptotics (uncertainty propagation through the extrapolation) are not developed here.

2. **Classical error.** The NHANES application makes the assumption auditable: the corrected estimates lie 1.5 standard errors *above* the measured-weight anchor, the signature of nonclassical (truth-correlated) self-report error, which no classical-error method removes. Users with validation data should always fit the anchor when it exists.

3. **Residual extrapolation bias at large error.** At $\sigma_\epsilon^2/\mathrm{Var}(X) \gtrsim 0.5$, polynomial SIMEX of any flavor retains bias of the same order as the naive attenuation it removes; coverage of nominal 95% intervals degrades accordingly. Remedies (deconvolution-aware selection; variance-penalized degree selection) are stated in Section **??** but not developed.

4. **First-order variance estimator.** The sandwich under-states the finite-sample variance by 10–34% at $n \leq 100$; bootstrap standard errors are the recommended small-sample alternative.

5. **High dimensions.** Repeated GLM fitting remains $O(np^2)$ per pseudo-sample; regularized SIMEX would be needed for p in the hundreds.

1.4 Extensions and Future Work

Beyond the Berkson error, imputation with validation data, alternative GLMs, time-series/spatial dependence, and regularized extensions sketched in the earlier draft (all still open), the present results add two concrete problems: (i) *deconvolution-aware model selection* — constructing a held-out criterion whose population minimizer is $\beta(-1) = \boldsymbol{\beta}_0$ rather than $\beta(1)$, e.g. by evaluating the held-out deviance at pseudo-deconvolved covariates simulated under each candidate hyperparameter — for which our CV-bias finding provides both the motivation and the diagnostic; and (ii) *variance-aware degree selection* — enlarging the degree grid while penalizing the extrapolation variance $\dot{\boldsymbol{C}}_\beta$ of the efficiency theorem, in the spirit of model-selection penalties, which would automate the bias–variance trade-off that practitioners currently negotiate by hand.

2 Conclusion

Measurement error in covariates is pervasive, and the naive Poisson fit is not merely inefficient but misleading: its attenuation grows linearly in the error variance and its intervals collapse, as our simulations document cell by cell. SIMEX remains the most practical correction framework for this problem, and this paper makes it automatic, gives it correct and complete asymptotics, equips it with a closed-form variance estimator, and reports—at replication standard, with all seeds fixed—exactly where it succeeds (small to moderate error, with valid inference and RMSE ties with the best competitor), what it costs (a quantified efficiency premium relative to the oracle), and where it fails (large error, where residual extrapolation bias dominates and observed-scale cross-validation under-extrapolates by construction). The NHANES application shows the workflow we recommend in practice: estimate the error variance from measurable discrepancies, correct, validate against any available anchor, and report the sensitivity profile over the error variance. The open-source implementation and the fully reproducible archive are offered so that both the method and the standard of evidence can be checked and extended.

References

Nakamura, T. (1990). *Biometrika*, 77(1), 127–137.

Delaigle, A., and Meister, A. (2008). *JASA*, 103(482), 869–880.